

AMENDMENT NO. 1 JUNE 2024
TO
IS 15888 : 2010 GUIDELINE FOR THE CONDUCT OF FOOD SAFETY
ASSESSMENT OF FOODS DERIVED FROM RECOMBINANT-DNA PLANTS

(Foreword, para 8, line 2) — Substitute ‘Annex A’ for ‘Annex’.

(Foreword, para 8) — Insert the following new para:

‘The Codex General Principles for the Addition of Essential Nutrients to Foods (CAC/GL 09-1987), generally applicable to the assessment of food derived from a plant which is modified by increasing the amount of a nutrient(s) or related substance(s) available for absorption and metabolism, was used in developing Annex B of these guidelines.’

(Page 6, clause 4.6.5.1) — Insert the following at the end:

‘A detailed presentation of issues to be considered is given in Annex B.’

(Page 8, clause 5.3) — Insert the following new clause:

‘5.4 Low-level Presence of Recombinant-DNA Material in Food

A detailed account of the food safety assessment in situation of low-level presence of recombinant-DNA material in food is given in Annex C.’

(Page 10, Annex A) — Insert the following new Annexures:

‘ANNEX B

(Clause 4.6.5.1)

FOOD SAFETY ASSESSMENT OF FOODS DERIVED FROM RECOMBINANT-DNA PLANTS
MODIFIED FOR NUTRITIONAL OR HEALTH BENEFITS

B-1 INTRODUCTION

B-1.1 This Annex provides additional considerations that are specific to foods modified for nutritional or health benefits. The document does not extend beyond a safety assessment and therefore, it does not cover assessment of the benefits themselves or any corresponding health claims, or risk-management measures.

B-1.2 The following factors determine whether a recombinant-DNA plant is a recombinant-DNA Plant Modified for Nutritional or Health Benefits, and as such within the scope of this Annex:

- a) The recombinant-DNA plant exhibits a particular trait in portion(s) of the plant intended for food use, and;
- b) The trait is a result of:
 - 1) introduction of a new nutrient(s) or related substance(s); or
 - 2) alteration of either the quantity or bioavailability of a nutrient(s) or related substance(s);
 - 3) removal or reduction of undesirable substance(s) (for example, allergens or toxicants), or; and
 - 4) alteration of the interaction(s) of nutritional or health relevance of these substances.

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B-2 DEFINITION

The definition below applies to this Annex:

B-2.1 Nutrient — Means any substance normally consumed as a constituent of food:

- a) which provides energy; or
- b) which is needed for growth and development and maintenance of healthy life, or; and
- c) a deficit of which will cause characteristic biochemical or physiological changes to occur.

B-3 FOOD SAFETY ASSESSMENT

B-3.1 The provisions of **3.3** and **4.6.5** applies to the overall safety assessment of a food derived from a recombinant-DNA plant modified for nutritional or health benefits. This Annex presents additional considerations regarding the food safety assessment of those foods.

B-3.2 Foods derived from recombinant-DNA plants modified for nutritional or health benefits may benefit certain populations/sub populations, while other populations/sub populations may be at risk from the same food.

B-3.3 Rather than trying to identify every hazard associated with a particular food, the intention of a safety assessment of food derived from recombinant-DNA plants is the identification of new or altered hazards relative to the conventional counterpart. Since recombinant-DNA plants modified for nutritional or health benefits result in food products with a composition that may be significantly different from their conventional counterparts, the choice of an appropriate comparator is of great importance for the safety assessment. Those alterations identified in a plant modified to obtain nutritional or health benefits are the subject of this safety assessment.

B-3.4 Upper levels of intake for many nutrients that have been set out by some national, regional and international bodies may be considered, as appropriate. The basis for their derivation should also be considered in order to assess the public health implications of exceeding these levels.

B-3.5 The safety assessment of related substances should follow a case-by-case approach taking into account upper levels as well as other values, where appropriate.

B-3.6 Although it is preferable to use a scientifically-determined upper level of intake of a specific nutrient or related substance, when no such value has been determined, consideration may be given to an established history of safe use for nutrients or related substances that are consumed in the diet if the expected or foreseeable exposure would be consistent with those historical safe levels.

B-3.7 With conventional fortification of food, typically a nutrient or a related substance is added at controlled concentrations and its chemical form is characterized. Levels of plant nutrients or related substances may vary in both conventionally bred and recombinant-DNA plants due to growing conditions. In addition, more than one chemical form of the nutrient might be expressed in the food as a result of the modification and these may not be characterized from a nutrition perspective. Where appropriate, information may be needed on the different chemical forms of the nutrient(s) or related substance(s) expressed in the portion of the plant intended for food use and their respective levels.

B-3.8 Bioavailability of the nutrient(s), related substance(s), or undesirable substance(s) in the food that were the subject of the modification in the recombinant-DNA plant should be established, where appropriate. If more than one chemical form of the nutrient(s) or related substance(s) is present, their combined bioavailability should be established, where appropriate.

B-3.9 Bioavailability will vary for different nutrients, and methods of testing for bioavailability should be relevant to the nutrient, and the food containing the nutrient, as well as the health, nutritional status and dietary practices of the specific populations consuming the food. In vitro and in vivo methods to determine bioavailability exist, the latter conducted in animals and in humans. In vitro methods can provide information to assess extent of release of a substance from plant tissues during the digestive process. In vivo studies in animals are of limited value in assessing nutritional value or nutrient bioavailability for humans and would require careful design in order to be relevant. In vivo studies, in particular, human studies may provide more relevant information about whether and to what extent the nutrient or related substance is bioavailable.

B-3.10 Guidance on dietary exposure assessment of foods derived from recombinant-DNA plants with nutritional modifications is provided in **4.6.5.2** of the standard. In the context of this Annex, dietary exposure assessment is the estimation of the concentration of the nutrient(s) or related substance(s) in a food, the expected or foreseeable consumption of that food, and any known factors that influence bioavailability. Exposure to a nutrient(s) or related substance(s) should be evaluated in the context of the total diet and the assessment should

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be carried out based on the customary dietary consumption, by the relevant population(s), of the corresponding food that is likely to be displaced. When evaluating the exposure, it is appropriate to consider information on whether the consumption of the modified food could lead to adverse nutritional effects as compared to consumption of the food that it is intended to replace. Most, if not all, aspects of exposure assessment are not unique to recombinant-DNA plants modified for nutritional or health benefits.

B-3.11 The first step of an exposure assessment is determining the level(s) of the substance(s) in question in the portion of the plant intended for food use. Guidance on determining changes in levels of these substances is provided in the standard.

B-3.12 Consumption patterns will vary from country to country depending on the importance of the food in the diet(s) of a given population(s). Therefore, it is recommended that consumption estimates are based on national or regional food consumption data when available, using existing guidance on estimation of exposure in a given population(s). Data on staple food products may also be supplemented by information from FAO Food Balance Sheets. When national or regional food consumption data is unavailable, food availability data may provide a useful resource.

B-3.13 To assess the safety of a food derived from a recombinant-DNA plant modified for a nutritional or health benefit, the estimated intake of the nutrient or related substance in the population(s) is compared with the nutritional or toxicological reference values, such as upper levels of intake, ADIs for that nutrient or related substance, where these values exist. This may involve assessments of different consumption scenarios against the relevant nutritional reference value, taking into account possible changes in bioavailability, or extend to probabilistic methods that characterize the distribution of exposures within the relevant population(s).

ANNEX C

(Clause 5.4)

**FOOD SAFETY ASSESSMENT IN SITUATIONS OF LOW-LEVEL PRESENCE OF
RECOMBINANT-DNA PLANT MATERIAL IN FOOD**

C-1 INTRODUCTION

C-1.1 An increasing number of recombinant-DNA plants are being authorized for commercialization. However, they are authorized at different rates in different countries. As a consequence of these asymmetric authorizations, low levels of recombinant DNA plant materials that have passed a food safety assessment according to the standard in one or more countries may on occasion be present in food in importing countries in which the food safety of the relevant recombinant-DNA plants has not been determined.

C-1.2 This Annex describes the recommended approach to the food safety assessment in such situations of low-level presence of recombinant-DNA plant material or in advance preparation for such potential circumstances. This guidance is not intended for a recombinant-DNA plant that was not authorized in an importing country as a result of that country's food safety assessment.

C-1.3 This Annex also describes data and information sharing mechanisms to facilitate utilization of the Annex and to determine whether it should apply.

C-1.4 This Annex can be applied in two different dietary exposure situations:

- a) That involving commodities, such as grains, beans or oil seeds, in which exposure to food from a variety not authorized in the importing country would likely be to dilute low level amounts at any one time. This would likely be the more common situation of low-level presence of recombinant-DNA plant material. Because any food serving of grains, beans or oil seeds would almost necessarily come from multiple plants, and because of how these types of commodities generally are sourced from multiple farms, are commingled in grain elevators, are further commingled in export shipments, at import and when used in processed foods, any inadvertently commingled material derived from

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recombinant-DNA plant varieties would be present only at a low level in any individual serving of food; and

- b) That involving foods that are commonly consumed whole and undiluted, such as some fruits and vegetables like potatoes, tomatoes, and papaya, in which exposure would be rare but could be to an undiluted form of the unauthorized recombinant-DNA plant material. While the likelihood of consuming material from such an unauthorized variety would be low and the likelihood of repeated consumption would be much lower, any such consumption might be of an entire unauthorized fruit or vegetable.

C-1.5 In both cases, the dietary exposure will be significantly lower than would be considered in a food safety assessment of the recombinant-DNA plant according to the standard. As a result, only certain elements of the standard will be relevant and therefore are included in this Annex.

C-1.6 This Annex does not:

- a) address risk management measures; national authorities will determine when a recombinant-DNA plant material is present at a level low enough for this Annex to be appropriate;
- b) preclude national authorities from conducting a safety assessment according to the standard; countries can decide when and how to use the Annex within the context of their regulatory systems, or; and
- c) eliminate the responsibility of industries, exporters and, when applicable, national competent authorities to continue to meet countries' relevant import requirements, including in relation to unauthorized recombinant-DNA plant material.

C-2 GENERAL AND OTHER CONSIDERATIONS

C-2.1 For the food safety assessment in situations of low-level presence of recombinant DNA plant materials in food, **4** and **5** of the standard apply as amended as follows. The applicable paragraphs are specifically indicated. Those paragraphs of the Codex Plant Guidelines that are not listed can be omitted from consideration.

C-2.2 Description of the Recombinant-DNA Plant

See 4.1 of the standard which is applicable here.

C-2.3 Description of the Host Plant and its Use as a Food

See 4.2.1, 4.2.2 and 4.2.3 of the standard which are applicable here.

C-2.4 Description of the Donor Organism(s)

Information should be provided on the donor organism(s) and, when appropriate, on other related species. It is particularly important to determine if the donor organism(s) or other closely related members of the family naturally exhibit characteristics of pathogenicity or toxin production, or have other traits that affect human health. The description of the donor organism(s) should include:

- a) its usual or common name;
- b) scientific name;
- c) taxonomic classification;
- d) information about the natural history as concerns food safety;
- e) information on naturally occurring toxins and allergens; for microorganisms, additional information on pathogenicity and the relationship to known pathogens; and,
- f) information on past and present use, if any, in the food supply and exposure route(s) other than intended food use (for example, possible presence as contaminants).

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C-2.5 Description of the Genetic Modification(s)

See 4.4.1, 4.4.2 and 4.4.3 of the standard which are applicable here.

C-2.6 Characterization of the Genetic Modification(s)

C-2.6.1 *See 4.5.1 and 4.5.2 of the standard which are applicable here.*

C-2.6.2 Information should be provided on any expressed substances in the recombinant-DNA plant; this should include:

- a) the gene product(s) (for example, a protein or an untranslated RNA);
- b) the gene product(s)' function;
- c) the phenotypic description of the new trait(s);
- d) the level and site of expression in the plant of the expressed gene product(s), and the levels of its metabolites in the edible portions of the plant; and
- e) where possible, the amount of the target gene product(s) if the function of the expressed sequence(s)/gene(s) is to alter the accumulation of a specific endogenous mRNA or protein.

C-2.6.3 *See 4.5.4 of the standard which is applicable here.*

C-2.7 Safety Assessment

C-2.7.1 Expressed Substances (Non-Nucleic Acid Substances)

C-2.7.1.1 Assessment of possible toxicity

C-2.7.1.1.1 The safety assessment should take into account the chemical nature and function of the newly expressed substance and identify the concentration of the substance in the edible parts of the recombinant-DNA plant, including variations and mean values.

C-2.7.1.1.2 Information should be provided to ensure that genes coding for known toxins present in the donor organisms are not transferred to recombinant-DNA plants that do not normally express those toxic characteristics. This assurance is particularly important in cases where a recombinant-DNA plant is processed differently from a donor plant, since conventional food processing techniques associated with the donor organisms may deactivate, degrade or eliminate toxicants.

C-2.7.1.1.3 *See 4.6.1.1.4 of the standard which is applicable here.*

C-2.7.1.1.4 In the case of proteins, the assessment of potential toxicity should focus on amino acid sequence similarity between the protein and known protein toxins as well as stability to heat or processing and to degradation in appropriate representative gastric and intestinal model systems. appropriate oral toxicity studies may need to be carried out in cases where the protein present in the food is not similar to proteins that have previously been consumed safely in food, and taking into account its biological function in the plant where known.

C-2.7.1.1.5 *See 4.6.1.1.6 and 4.6.1.1.7 of the standard which are applicable here.*

C-2.7.1.2 Assessment of possible allergenicity (proteins)

See 4.6.1.2.1, 4.6.1.2.2 and 4.6.1.2.3 of the standard which are applicable here.

C-2.7.1.3 Analyses of key toxicants and allergens

C-2.7.1.3.1 Key toxicants are those toxicologically significant compounds known to be inherently present in the plant, such as those compounds whose toxic potency and level may be significant to health (for example, solanine in potatoes if the level is increased). Analyses of key toxicants and allergens are important in certain cases of foods from recombinant-DNA plants (for example, those that are commonly consumed whole and undiluted, such

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as potatoes, tomatoes, and papaya). Analyses of concentrations of key toxicants and allergens of the recombinant-DNA plant typical of the food should be compared with an equivalent analysis of a conventional counterpart grown and harvested under the same conditions. The statistical significance of any observed differences should be assessed in the context of the range of natural variations for that parameter to determine its biological significance. The comparator(s) used in this assessment should ideally be the near isogenic parental line. In practice, this may not be feasible at all times, in which case a line as close as possible should be chosen. The purpose of this comparison is to establish that substances that can affect the safety of the food have not been altered in a manner that would have an adverse impact on human health.

C-2.7.1.3.2 The location of trial sites should be representative of the range of environmental conditions under which the plant varieties would be expected to be grown. The number of trial sites should be sufficient to allow accurate assessment of key toxicants and allergens over this range. Similarly, trials should be conducted over a sufficient number of generations to allow adequate exposure to the variety of conditions met in nature. To minimize environmental effects, and to reduce any effect from naturally occurring genotypic variation within a crop variety, each trial site should be replicated. An adequate number of plants should be sampled and the methods of analysis should be sufficiently sensitive and specific to detect variations in key toxicants and allergens.

C-2.7.1.4 *Evaluation of metabolites*

Some recombinant-DNA plants may have been modified in a manner that could result in new or altered levels of various metabolites in the food. In certain cases of foods from recombinant-DNA plants (for example, those that are commonly consumed whole and undiluted), consideration should be given to the potential for the accumulation of metabolites in the food that would adversely affect human health. Food safety assessment in situations of low level presence of recombinant-DNA material in foods from such plants requires investigation of residue and metabolite levels in the food. Where altered residue or metabolite levels are identified in foods, consideration should be given to the potential impacts on human health using conventional procedures for establishing the safety of such metabolites (for example, procedures for assessing the human safety of chemicals in foods).

C-2.7.1.5 *Food processing*

The potential effects of food processing, including home preparation, on foods derived from recombinant-DNA plants should also be considered. For example, alterations could occur in the heat stability of an endogenous toxicant. Information should therefore be provided describing the processing conditions used in the production of a food ingredient from the plant. For example, in the case of vegetable oil, information should be provided on the extraction process and any subsequent refining steps.

C-2.7.1.6 *Potential accumulation of substances significant to human health*

Some recombinant-DNA plants may exhibit traits (for example, herbicide tolerance) which may indirectly result in the potential for accumulation of pesticide residues, altered metabolites of such residues, toxic metabolites, contaminants, or other substances which may be relevant to human health. In certain cases of foods from recombinant-DNA plants (for example, those that are commonly consumed whole and undiluted), the risk assessment should take this potential for accumulation into account. Conventional procedures for establishing the safety of such compounds (for example, procedures for assessing the human safety of chemicals) should be applied.

C-2.7.1.7 *Use of antibiotic resistance marker genes*

See 5.2.1, 5.2.2, 5.2.3 and 5.2.4 of the standard which are applicable here.

C-3 GUIDANCE ON DATA AND INFORMATION SHARING

C-3.1 In order to use this Annex, it is essential to have access to requisite data and information.

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C-3.2 Relevant agencies of the country should make available to a publicly accessible central database to be maintained by FAO information on recombinant-DNA plants authorized in accordance with the standard. This information should be presented in accordance with the following format:

- a) Name of product applicant;
- b) Summary of application;
- c) Country of authorization;
- d) Date of authorization;
- e) Scope of authorization;
- f) Unique identifier;
- g) Links to the information on the same product in other databases maintained by relevant international organizations, as appropriate;
- h) Summary of the safety assessment, which should be consistent with the framework of food safety assessment of the standard;
- j) Where detection method protocols and appropriate reference material (non-viable, or in certain circumstances, viable) suitable for low-level situation may be obtained. The information may be provided by the product applicant or in some cases by relevant agencies of the country; and
- k) Contact details of the competent authority(s) responsible for the safety assessment and the product applicant.

C-3.3 This process should facilitate rapid access by importing relevant agencies of the country to additional information relevant to the assessment of food safety assessment in situations of low-level presence of recombinant-DNA plant material in foods in accordance with this Annex.

C-3.4 The authorizing relevant agencies of the country should make available complementary information to other agencies of the country on its safety assessment in accordance with the standard, in conformity with its regulatory/legal framework.

C-3.5 The product applicant should provide further information and clarification as necessary to allow the assessment according to this Annex to proceed, as well as a validated protocol for an event-specific or trait-specific detection method suitable for low level situations and appropriate reference materials (non-viable, or in certain circumstances, viable). This is without prejudice to legitimate concerns to safeguard the confidentiality of commercial and industrial information.

C-3.6 As appropriate, new scientific information relevant to the conclusions of the food safety assessment conducted in accordance with the standard by the authorizing agencies of the country should be made available.'

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